

Nathaniel Lee

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EDUCATION

University of Chicago

B.S. Computer Science, B.A. Statistics

GPA: 3.83/4.00

Chicago, IL

Expected: June 2028

Relevant Coursework: Distributed Systems, Math for Machine Learning, Data Structures and Algorithms, Discrete Mathematics, Linear Algebra, Differential Equations, System Programming, Intro to Probability, Software Development

EXPERIENCE

Naive (YC X25, \$28.5M Series A)

Early Engineering Hire

March 2026 – August 2026

San Francisco, CA

- Shipped the production container runtime for all customer AI agents (**30K users** at launch) with the founding engineering team, securely isolating each customer's agents and credentials without downtime.
- Eliminated over **90%** of billed VM time per session by routing 83% of agent commands into a lightweight sandbox and creating VMs only on demand.
- Cut per-customer infrastructure cost **67%** by packing three isolated customer containers onto each VM instead of running one VM per customer.
- Cut inference cost **22%** and input tokens **40%** on the SWE-bench Pro coding benchmark by routing file exploration to a cheap scout model and reserving the frontier model for the fix, with no accuracy loss.

Computer Science Instructional Laboratory

Software Engineer Intern

June 2025 – Present

Chicago, IL

- Reached **50 active users** across 7 teams with an AI documentation platform (Python, FastAPI, Slack/Trello APIs) that assembles a task's documentation from tickets and docs scattered across tools.
- Cut Trello card verification time **40%** by integrating gpt-4o-mini (migrated from self-hosted LLaMA 3 on Ollama) with a React/Next.js frontend.
- Dropped follow-up questions **75%** by building a hybrid retrieval layer over related documents, combining vector ranking with 2-hop graph traversal.

PROJECTS

MisoTTS Inference Optimizer | *real-time inference optimization* | Python, PyTorch | GitHub

June 2026

- Accelerated an 8B text-to-speech model to real-time streaming speed (real-time factor 4.0→0.98, **4.1×**) on an A100 by profiling the inference pipeline, batching kernel launches, and reducing idle GPU time.
- Doubled GPU utilization (**27% to 56%**) by collapsing 219,627 kernel launches into 86,913, holding memory flat at 18.1 GB on the same hardware.

Slipstream | *cross-user action memory for web agents* | Python, vector search | GitHub

March 2026

- Won 3rd place and Best Real-Time Data at the YC Browser Use hackathon by building shared cross-agent memory that cut average task time **49%**.
- Cut steps per task up to **44%** on repeat runs by scoring every attempt and storing only the highest-scoring path for reuse.

Dojo | *GRPO post-training with held-out evaluation* | Python, PyTorch, Modal

July 2026

- Built a GRPO teacher to generate RL arenas, producing a downstream PPO student that beat the base-model baseline across **218 held-out evaluations**.

ACTIVITIES/LEADERSHIP

UChicago APEX Labs

Research Assistant

December 2025 – March 2026

Chicago, IL

- Built an automated two-pass OCR correction pipeline for a scanned-document archive, replacing page-by-page manual proofreading; **96%** of 1,410 pages reached ≥ 0.99 similarity to the human reference.
- Designed a confidence-scoring pass that flags uncertain pages for human review, validated at **84%** agreement between independent reviewers across 46K flagged errors.

TECHNICAL SKILLS

Languages: Python, C, Go, Java, TypeScript

AI / ML: RAG (Embeddings, Vector Search), MCP, LoRA Fine-Tuning, PyTorch, NumPy

Infrastructure & Tools: AWS (ECS/Fargate, S3, ECR), Docker, Terraform, Linux, Git, PostgreSQL

Hobbies: Hiking, Boulderling, Surfing, Hackathons (HackHarvard, YC Browser Use)